

AMD Ryzen Threadripper 9980X

Extreme desktop power for creators

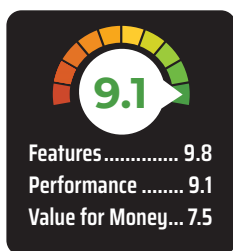
For years, AMD's Threadripper series has been synonymous with brute-force multi-core power. With the new AMD Ryzen Threadripper 9980X, AMD is sharpening that edge even further, bringing 64 cores and 128 threads to the TRX50 platform for high-end desktop users who demand no compromises. Built on the latest Zen 4 architecture and manufactured on TSMC's 5nm process, the 9980X is part of the Ryzen Threadripper 9000 series, aimed squarely at professional creators, advanced developers, and anyone juggling massive workloads, be it AI model training, 3D rendering, video production, or virtualised environments.

In many ways, this chip represents a reassertion of dominance in a niche that AMD essentially owns. The 9980X is not meant for the mainstream gamer or casual user. It's a product built for workstations that blur the line between desktop and server.

AMD RYZEN THREADRIPPER 9000 + TRX50 PLATFORM

The AMD Ryzen Threadripper 9980X is designed to run exclusively on the TRX50 platform, which forms the foundation for AMD's high-end desktop (HEDT) ecosystem. TRX50 isn't new, but it has matured significantly since the launch of the 7000 series, and with the 9000 series, it unlocks an impressive combination of bandwidth, overclocking flexibility, and workstation-grade features that blur the line between HEDT and server-class computing.

At the heart of TRX50 is the sTR5 socket, which is shared with the more enterprise-oriented WRX90 platform used by the PRO WX series. However,



there's a key distinction: while Threadripper PRO 9000 processors can run on both WRX90 and TRX50 boards, non-PRO chips like the 9980X are compatible only with TRX 50. This distinction is crucial for those seeking raw perfor-

mance without the cost or complexity of manageability and remote provisioning features found in enterprise platforms.

Memory support on TRX50 is robust. The platform handles quad-channel DDR5 RDIMM, with support for capacities of up to 1TB using four 256GB modules. This provides plenty of headroom for creators and developers dealing with

memory-hungry applications like virtualisation stacks, large-scale simulations, and high-resolution video editing workflows. The 9980X supports DDR5 memory speeds up to 6400 MT/s with EXPO profiles, and thanks to the platform's 1:2 memory controller mode, these speeds are achievable without manual tuning.

Expansion is another strength. The TRX50 platform offers a total of 92 PCIe lanes, with 80 of them supporting PCIe 5.0 directly from the CPU. This enables multi-GPU configurations, high-speed storage arrays, and professional-grade I/O cards to operate at full bandwidth without compromise. Whether you're building a data science workstation, a VFX rendering node, or an editing and inference rig, this level of connectivity gives the system enormous versatility.

THREADRIPPER TEST-RIG CONFIGURATION

To evaluate the real-world performance of the AMD Ryzen Threadripper 9980X, the test system was configured to eliminate any bottlenecks and reflect optimal conditions for high-end desktop workloads. AMD's guidelines were followed closely, with the latest drivers, BIOS, and firmware updates applied.

- **Processor:** The AMD Ryzen Threadripper 9980X
- **Cooler:** SilverStone XE360-TR5 (360 mm AIO)
- **Graphics card:** The NVIDIA GeForce RTX 5090
- **GPU driver:** NVIDIA GeForce 577.0
- **Motherboard:** GIGABYTE TRX50 AERO D

